

Procedure for Booting from an External Drive

1. **Perform a Full Shutdown:** Power down the machine completely. Avoid using sleep or hibernation states, as they keep the system kernel loaded and prevent a clean hardware initialization.
2. **Attach the External Media:** Plug your bootable USB flash drive or alternative external drive into an available port.
3. **Initiate the Startup Sequence:** Turn the computer on and immediately start tapping the appropriate manufacturer boot key repeatedly the moment the splash screen appears.
4. **Choose the Target Drive:** Once the boot selection window opens, navigate using the arrow keys to highlight your external device, then press Enter.

Troubleshooting Note: If the external drive fails to appear in the list, enter the system BIOS utility during startup using your model's specific setup key. Go to the Boot or Security settings panel, turn off Secure Boot, and activate Legacy Boot, CSM, or the F12 Boot Menu option before trying again.

Universal AMD64 Boot Menu & BIOS Key Reference

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Acer	Aspire, TravelMate, Predator, Veriton, Extensa	F12, Esc	F2, Del
AMD (Reference)	Evaluation Boards, Partner Reference Platforms	F12, F11	Del, F2

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Apple (Intel-based)	iMac, MacBook Pro, MacBook Air, Mac mini, Mac Pro	Option (Alt)	Command + Option + P + R (NVRAM) or hold power for firmware
ASRock	Desktop Motherboards (Intel & AMD Chipsets)	F11	F2, Del
ASUS	Zenbook, VivoBook, ROG, TUF, ProArt, Desktop PCs	Esc, F8	F2, Del
Atmel / AVR (Reference)	Embedded x86 Development Platforms	F12	Del
Biostar	Desktop Motherboards	F9, F12	Del, F2
Bull / Evidian	Enterprise Servers & Business Workstations	F11	F2, Del
Clevo	OEM Barebone Laptops (Sager, Prostar, etc.)	F12	F2, Del
Compaq	Presario, Evo, ProLiant Series	Esc, F9	F10
Dell	Inspiron, Latitude, OptiPlex, XPS, Precision, Alienware	F12	F2

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
DFI	Industrial & LanParty Motherboards	F8, F11	Del
ECS (Elitegroup)	Desktop Motherboards & Laptops	F11, F12	Del, F2
eMachines	Desktops, Laptops, Netbooks	F12	Tab, Del
EpoX	Legacy Desktop Motherboards	F8, F12	Del
EVGA	Desktop Motherboards (Intel & AMD Chipsets)	F11, F12	Del
FIC	First International Computer Motherboards	F12	Del, F2
Foxconn	OEM Motherboards & Desktop Systems	F11, F12	Del, F2
Fujitsu	LifeBook, Esprimo, Celsius Workstations	F12	F2
Gateway	NV Series, NE Series, FX Desktops	F11, Esc, F12	F2, Del
Getac	Fully Rugged Laptops & Mobile Handhelds	F12	F2, Del

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Gigabyte / AORUS	Desktop Motherboards (Intel & AMD Chipsets)	F12	Del
HP (Hewlett-Packard)	Pavilion, Envy, EliteBook, ProBook, OMEN, Z Workstations	Esc (then F9)	Esc (then F10)
IBASE	Industrial Motherboards & Embedded Systems	F11, F12	Del, F2
Intel	Desktop Boards (Legacy D-series, NUCs)	F10	F2
Jetway	Industrial & Consumer Motherboards	F11, F12	Del, F2
Lenovo	ThinkPad, IdeaPad, Legion, Yoga, ThinkCentre, ThinkStation	F12, F8, Fn+F11	F1, F2
LG	Gram Laptops, All-in-One PCs	F10, F12	F2
Maxdata	Business Laptops and Desktops	F12	F2, Del
Medion	Consumer Laptops, Desktops (ALDI brand)	F12	F2, Del

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Micro-Star Int. (MSI)	Gaming Laptops, Creator Series, Motherboards	F11	Del
Mitac	Enterprise Servers & OEM Laptops	F11, F12	Del, F2
NEC	Versa, LaVie Series	F12	F2
Packard Bell	Laptops, EasyNote, Desktops	F12, F8	F2, Del
Panasonic	Toughbook Rugged Laptops & Tablets	F12	F2, Del
PC Partner	OEM Motherboards & Video Graphics Platforms	F11, F12	Del
Portwell	Industrial SBCs & Embedded Computers	F11, F12	Del, F2
QNAP	NAS Systems (x86 architecture variants)	F11, F12	Del, F2
Quanta Computer	OEM Laptop Architecture	F12	F2, Del
Razer	Razer Blade Laptops	F12	F1, F2

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Samsung	Ativ Book, Notebook 9, Odyssey, Series 3/5/7/9	Esc, F12	F2
Shuttle	XPC Mini PCs & Barebone Systems	F8, F12	Del
Sony	VAIO (Duo, Pro, Flip, Fit, Tap, older VGN/PCG series)	F11, or ASSIST button	F1, F2, F3
Supermicro	Server Boards, Workstations, AMD EPYC/Intel Xeon systems	F11	Del
System76	Linux Laptops, Desktops, Servers (Galago, Lemur, etc.)	F12	F2, Del
Tyan	Server & Workstation Motherboards	F11, F12	Del, F2
Toshiba / Dynabook	Satellite, Portege, Tecra, Equium	F12	F1, Esc
Twinhead	Durabook Rugged Laptops	F12	F2, Del

Manufacturer / Brand	Target System Types & Motherboards	Boot Menu Key	BIOS / Setup Key
Unisys	Enterprise Servers & Mainframe-adjacent x86 systems	F11, F12	Del, F2
VIA Technologies	Mini-ITX Embedded Motherboards	F11, F12	Del, F2
ViewSonic	Smart Displays & All-in-One Touch PCs	F12	F2, Del
Wistron	OEM Laptop Manufacturing Architecture	F12	F2, Del
Xen / KVM / Virtual	Virtual Machines (QEMU, Proxmox, oVirt, OpenStack)	Esc, F12	F2, Del
Xerox	Digital Printing Press Controllers & Workstations	F11, F12	Del, F2
Zotac	ZBOX Mini PCs	F8, F12	Del